

18 Electric fields

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Electric field strength E is force per unit positive charge. Between parallel plates, uniform $E = V/d$; a point charge produces a radial field proportional to $1/r^2$. Electric potential is energy per unit charge. A positive charge tends to move in the direction of decreasing potential when free.

Worked example: follow the reasoning

A 1200 V p.d. across plates separated by 0.020 m produces $E = 6.0 \times 10^4 \text{ V m}^{-1}$. A proton feels $F = eE \approx 9.6 \times 10^{-15} \text{ N}$.

Quick check · multiple choice

A Between uniform parallel plates, E equals what? [1]

- A Vd
- B V/d
- C d/V
- D V^2d

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 Parallel plates 5.0 mm apart have p.d. 300 V. Find field strength and work done moving an electron across the p.d. [3]

Final answer: _____

Answers and explanation

Quick check: B. V/d

Question 1 [3 marks]

$E = V/d = 300/0.005 = 6.0 \times 10^4 \text{ V m}^{-1}$ [2]; energy magnitude $eV = 1.60 \times 10^{-19} \times 300 = 4.8 \times 10^{-17} \text{ J}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.